Summary Notes

Earners AI and ML by Aditya Jain sir - lecture 10

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Recap

```
for i in range(5):
  for j in range(6,10):
    print(i,j)
```

```
0 6
0 7
0 8
0 9
1 6
1 7
1 8
1 9
2 6
2 7
2 8
2 9
3 6
3 7
3 8
3 9
4 6
4 7
4 8
4 9
```

HW of Last Lecture

```
What is the output?
for i in range(2):
    for j in range(2):
        print(j, end=" ")
    print()

A.

0 1
0 1

B.

1 2
1 2

C.

0
1

D. Error

Ans: A
```

```
  File "/tmp/ipykernel_23922/3765884299.py", line 7
    A.
      ^
SyntaxError: invalid syntax
```

Next steps: Explain error

```
for i in range(2):
    for j in range(2):
        print(j, end=" ")
    print()
```

```
0 1
0 1
```

```
Which pattern will this code print?
for row in range(4):
    for col in range(4):
        print("#", end=" ")
    print()

A.

#
##
###
####

B.

# # # #
# # # #
# # # #
# # # #

C.

####
####
####
####

D.

#
#
#
#
```

```
for row in range(4):
    for col in range(4):
        print("#", end=" ")
    print()
```

```
# # # #
# # # #
# # # #
# # # #
```

```
What is the output?
for i in range(1,3):
    for j in range(1,3):
        print(i + j, end=" ")
    print()

A.

2 3
3 4

B.

1 2
2 3

C.

2 4
4 6

D.

3 5
```

```
  File "/tmp/ipykernel_23922/2337426660.py", line 7
    A.
      ^
SyntaxError: invalid syntax
```

Next steps: Explain error

```
for i in range(1,3):
    for j in range(1,3):
        print(i,j,i + j)
    print()
```

```
1 1 2
1 2 3

2 1 3
2 2 4
```

List

```
m1 = 90
m2 = 86
m3 = 95
m4 = 83
m5 = 70
```

```
marks = [90, 86, 95, 83, 70]
```

```
print(marks)
```

```
[90, 86, 95, 83, 70]
```

```
a = [45, "AJ", 6.8, True]
```

```
print(a)
```

```
[45, 'AJ', 6.8, True]
```

```
type(a)
```

```
list
```

```
type(marks)
```

```
list
```

```
students = []
print(students)
```

```
[]
```

Indexing

```
#indexing starts at 0
print(marks)
```

```
[90, 86, 95, 83, 70]
```

```
marks[0]
```

```
90
```

```
marks[2]
```

```
95
```

```
marks[4]
```

```
70
```

```
n = len(marks)
```

```
print(n)
```

```
5
```

```
marks[n-1]
```

```
70
```

```
marks[n-2]
```

```
83
```

```
marks[-1]
```

```
70
```

```
marks[-2]
```

```
83
```

```
marks
```

```
[90, 86, 95, 83, 70]
```

```
marks[5]
```

```
---------------------------------------------------------------------------
IndexError                                Traceback (most recent call last)
/tmp/ipykernel_23922/2760646083.py in <cell line: 0>()
----> 1 marks[5]

IndexError: list index out of range
```

Next steps: Explain error

```
marks
```

```
[90, 86, 95, 83, 70]
```

```
marks[2] = 97
```

```
marks
```

```
[90, 86, 97, 83, 70]
```

```
# marks(2) = 97
```

Iterating over a list

```
n = len(marks)
print(n)
```

```
5
```

```
marks[0]
```

```
90
```

```
marks[1]
```

```
86
```

```
for i in range(n):
  print(marks[i])
```

```
90
86
97
83
70
```

```
for elem in marks:
  print(elem)
```

```
90
86
97
83
70
```

```
marks
```

```
[90, 86, 97, 83, 70]
```

```
#Write a code to print the elements present at even indexes
for i in range(n):
  if i%2 == 0:
    print(marks[i])
```

```
90
97
70
```

```
#Write a code to print all the even elements

for i in range(n):
  if marks[i]%2 == 0:
    print(marks[i])
```

```
90
86
70
```

```
for elem in marks:
  if elem%2 == 0:
    print(elem)
```

```
90
86
70
```

Quiz

```
What is the output?

numbers = [10, 20, 30]
print(numbers[0])
```

```
Object `output` not found.
10
```

```
numbers = [10, 20, 30]
print(numbers[0])
```

```
10
```

```
# What is the output?

fruits = ["Apple", "Banana", "Mango"]
print(fruits[-1])
```

```
Mango
```

```
# What is the output?

marks = [70, 80, 90]
print(len(marks))
```

```
3
```

```
Which of the following is a valid Python list?

A.

marks = (10,20,30)

B.

marks = {10,20,30}

C.

marks = [10,20,30]

D.

marks = <10,20,30>

ans : C
```

```
  File "/tmp/ipykernel_23922/21065083.py", line 3
    A.
      ^
SyntaxError: invalid syntax
```

Next steps: Explain error

```
# What is the output?
numbers = [5,10,15,20]

print(numbers[-2])
```

```
15
```

```
# What is the output?

fruits = ["Apple","Banana","Mango"]

fruits[1] = "Orange"
```

```
print(fruits)
```

```
['Apple', 'Orange', 'Mango']
```

```
# What is the output?

numbers = [10,20]

print(numbers[2])
```

```
---------------------------------------------------------------------------
IndexError                                Traceback (most recent call last)
/tmp/ipykernel_23922/3036336073.py in <cell line: 0>()
      3 numbers = [10,20]
      4
----> 5 print(numbers[2])

IndexError: list index out of range
```

Next steps: Explain error

```
# What is the output?

numbers = [1,2,3]

for num in numbers:
    print(num)
```

```
1
2
3
```

```
# What is printed?

numbers = [5,10,15]

for i in range(len(numbers)):
    print(numbers[i])
```

```
5
10
15
```

```
# What is the output?


marks = [60,70,80,90]

print(max(marks))
```

```
90
```

```
marks = [60,70,80,90]

print(min(marks))
```

```
60
```

```
#Write a code to find the sum of all the elements of the list.

marks = [60,70,80,90]

sm = 0
for elem in marks:
  sm = sm + elem

print(sm)
```

```
300
```

```
marks = [60,70,80,90]
```

```
sum(marks)
```

```
300
```

```
#Write a code to find the average  of all the elements of the list.
```

```
marks = [60,70,80,90]
n = len(marks)
sm = 0
for elem in marks:
  sm = sm + elem

print(sm/n)
```

```
75.0
```

```
# What is printed?

numbers = [9,5,7,1]

print(sorted(numbers))
```

```
[1, 5, 7, 9]
```

```
numbers
```

```
[9, 5, 7, 1]
```

```
numbers.sort()
```

```
print(numbers)
```

```
[1, 5, 7, 9]
```

```
# What is the output?

names = ["A","B","C"]

print(names[len(names)-1])
```

```
C
```

```
numbers = [5, 10, 15]

numbers[1] = numbers[0] + numbers[2]

print(numbers)

print(sum(numbers))
```

```
[5, 20, 15]
40
```

```
marks = [50, 60, 70, 80]

for i in range(len(marks)):
    marks[i] = marks[i] + 5

print(marks)
```

```
[55, 65, 75, 85]
```

```
data = [2, 4, 6]

for i in range(len(data)):
    print(i, data[i])
```

```
0 2
1 4
2 6
```

```
numbers = [8, 2, 5, 1]

print(min(numbers))

print(max(numbers))

print(max(numbers) - min(numbers))
```

```
1
8
7
```

```
items = [1, 2, 3]

for i in range(len(items)):
    items[i] = items[i] * 2

print(sum(items))
```

```
12
```

```
numbers = [10, 20, 30]

print(numbers[-1])

numbers[-1] = 100

print(numbers)
```

```
30
[10, 20, 100]
```

```
nums = [2, 4, 6, 8]

count = 0

for n in nums:
    if n > 4:
        count += 1

print(count)
```

```
2
```

```
marks = [80, 75, 90]

highest = max(marks)

for m in marks:
    print(highest - m)
```

```
10
15
0
```

```
numbers = [3, 5, 7]

result = 1

for n in numbers:
    result = result * n

print(result)
```

```
105
```

```
values = [4, 8, 12]

i = 0

while i < len(values):
    print(values[i] // 2)
    i += 1
```

```
2
4
6
```

HW

```
numbers = [1, 2, 3, 4]

for i in range(len(numbers)):
    numbers[i] = numbers[i] + i

print(numbers)
```

```
marks = [65, 75, 85]

total = 0

for m in marks:
    if m >= 70:
        total += m

print(total)
```

```
nums = [2, 4, 6]

nums[0] = nums[2]

nums[2] = nums[1]

print(nums)
```

```
names = ["Amit", "Neha", "Raj"]

longest = names[0]

for name in names:
    if len(name) > len(longest):
        longest = name

print(longest)
```

```
data = [10, 20, 30]

for i in range(len(data)):
    data[i] = data[i] + 10

print(max(data))

print(min(data))
```

```
numbers = [5, 10, 15, 20]

total = 0
```